

Agent-Based Simulation of Queue Dynamics: The Role of Personality and Emotional Stress

Word Count: 1018 (reference list not included)

Part 1: Partial ODD Description

1.1 Purpose and Patterns

The primary aim of this agent-based model is to simulate and analyse human queueing behaviour in a structured environment, such as a café or retail counter. Specifically, the model investigates how different personality types—**patient**, **assertive**, and **rule-breaker**—interact within a single queueing system, and how these interactions affect waiting times, emotional responses (anger, impatience), and overall service efficiency. Key research questions include:

- How do the proportions of personality types influence queue dynamics?
- What are the impacts of queue-jumping behaviours on waiting time and collective anger?
- Under what conditions do individuals leave the queue prematurely due to frustration or impatience?
- How does the system cope with different mixes of assertive and disruptive behaviours?

Patterns of interest include the emergent structure of the queue, periodic fluctuations in queue length, and the cumulative emotional load on the system, as well as patterns of abandonment (rage quits) triggered by system stressors.

1.2 Entities, State Variables, and Scales

Entities: The primary agents are *turtles*, representing customers, each endowed with a distinct personality type. The *environment* consists of a spatially embedded store with a service counter and entrance points.

Turtles (Agents) have the following **state variables**:

- personality (string): One of "patient", "assertive", or "rule-breaker".
- waiting-time (integer): Time steps spent in the queue.
- anger (integer): Increases when cut in line.
- impatience (integer): Increases over time regardless of events.
- in-queue? (boolean): Whether the agent is currently in the queue.

Globals include:

- queue-line: The ordered list of queued agents.
- served-count: Number of agents served.
- Counters for each personality's total count, cumulative waiting time, total anger, and rage quit events.

The model operates on a 2D grid representing the store layout. The spatial scale is abstract, with patches standing in for queue positions and walls.

1.3 Process Overview and Scheduling

Each tick in the simulation represents a discrete time step. The following processes occur in order:

1. Turtle Activation:

- Agents that are not in the queue attempt to join it based on their personality:
 - *Patient* agents join the back.
 - *Assertive* agents merge at the midpoint.
 - *Rule-breakers* push to the front, displacing others.
- Each agent increments its waiting-time and impatience.
- If anger or impatience exceeds predefined thresholds, the agent leaves the queue and is removed from the simulation.

2. Serving Customers:

- Every 10 ticks, the first agent in the queue is served (removed), and their waiting time is recorded.
- Queue positions are updated accordingly.

3. Replacement:

- If the total number of turtles falls below 30, new agents are spawned at the entrance with randomly assigned personalities based on adjustable probabilities.

1.5 Initialization

Upon setup:

- All global counters are reset.
- The queue is emptied.
- 30 agents are created and randomly positioned near one of two entrances.
- The agent's anger and impatience tolerance are set by sliders.
- Each agent is assigned a personality according to user-defined probabilities via sliders.
- Each personality is associated with a colour for visual clarity (blue for patient, orange for assertive, red for rule-breaker).

The store environment, including the counter, walls, and entrances, is drawn on the patch grid.

1.6 Input Data

This model does not rely on external datasets. All input parameters—such as the personality distribution probabilities (prob-patient, prob-rule-breaker), and thresholds for anger and impatience—are defined within the Net-Logo interface through sliders.

1.7 Submodels

Queueing Submodels:

- join-back-of-queue: Assigns the turtle to the end of the queue.
- try-to-merge-queue: Inserts the agent in the middle, shifting others and increasing their anger.
- push-to-front: Inserts the agent at the front, maximizing disruption and anger

response.

Emotional Dynamics:

- **update-waiting-time:** Increments both waiting time and impatience.
- **shift-back-from:** Moves agents backward when the queue is disrupted, increasing anger based on proximity to the front.
- **leave-queue:** Removes an agent due to excessive emotional stress and records cumulative statistics.

Service and Replenishment:

- **serve-next-customer:** Handles the removal of the frontmost turtle at scheduled intervals.
 - **create-new-turtle:** Maintains constant simulation pressure by spawning replacements at entrance patches.
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Part 2: Situation of Model Within the Literature

Queueing behavior has long been a subject of interest in both operations research and agent-based modeling (ABM), particularly in domains such as retail, transport hubs, and public services. This model contributes to a growing body of ABMs that explore how individual traits and decision-making heuristics shape emergent crowd dynamics, often in contrast to classical queueing theory which assumes homogeneous, rational agents.

In terms of system representation, the model reflects themes from established works on pedestrian flow and social norms in queues, such as Helbing et al. (2000), who introduced social force models to simulate pedestrian interactions, and Schelling-like segregation models (Schelling, 1971), where minor rule deviations lead to large systemic changes. Here, individual personality traits drive discrete spatial behavior, challenging the normative assumption of a first-come, first-served queue.

From a **design concept** standpoint, the model utilizes:

- **Emergence:** Queue dynamics, such as bottlenecks or rage-quits, arise from decentralized agent interactions, aligning with Epstein (1999), who emphasized the value of ABMs in capturing emergent macro-level outcomes from micro-level rules.
- **Heterogeneity:** Different personalities lead to diverging strategies and emotional responses, reflecting Bonabeau's (2002) emphasis on heterogeneity as a fundamental aspect of realistic agent-based systems.
- **Stochasticity:** Entry points and personality types are assigned probabilistically, introducing randomness to arrival patterns and contributing to systemic variability.
- **Adaptation and Feedback:** While agents do not learn over time, they react emotionally to disruptive events such as being cut in line. This form of reactive adaptation fits within Conte and Paolucci's (2001) framework of socially situated agents, where local interactions produce socially meaningful outcomes.
- **Sensing and Interaction:** Agents assess their position and respond to queue

disruptions in real time, simulating embodied, bounded rational behavior under local constraints (Dai, Yu and Wu, 2024).

This model thus bridges micro-level psychological traits and macro-level service outcomes. It advances the literature by explicitly incorporating emotion-driven decision making, such as anger and impatience—as key behavioral drivers. This complement works like Moussaïd et al. (2011), which link individual cognitive and affective responses with crowd behavior in high-stress environments.

In summary, the model provides a novel ABM framework for exploring fairness, disruption, and behavioral diversity in structured queueing systems, contributing to the broader literature on social simulation and crowd dynamics.

Reference list

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